rather than their positions and velocities as in our earlier example—are correlated. The details of how this is done are not important; what is important is that it can be done and, in fact, can be done easily. The correlation can be arranged so that if the left and right detectors are set to measure the spins along axes pointing in the same direction, they will get the same result: if the detectors are set to measure the spin of their respective incoming electrons about a vertical axis and the left detector finds that the spin is clockwise, so will the right detector; if the detectors are set to measure spin along an axis 60 degrees clockwise from the vertical and the left detector measures a counterclockwise spin, so will the right detector; and so on. Again, in quantum mechanics the best we can do is predict the probability that the detectors will find clockwise or counterclockwise spin, but we can predict with 100 percent certainty that whatever one detector finds the other will find, too.[7]

Bohm's refinement of the EPR argument is now, for all intents and purposes, the same as it was in the original version that focused on position and velocity. The correlation between the particles' spins allows us to measure indirectly the spin of the left-moving particle about some axis by measuring that of its right-moving companion about that axis. Since this measurement is done far on the right side of the laboratory, it can't possibly influence the left-moving particle in any way. Hence, the latter must all along have had the spin value just determined; all we did was measure it, albeit indirectly. Moreover, since we could have chosen to perform this measurement about *any* axis, the same conclusion must hold for any axis: the left-moving electron must have a definite spin about each and every axis, even though we can explicitly determine it only about one axis at a time. Of course, the roles of left and right can be reversed, leading to the conclusion that each particle has a definite spin about any axis.[12]

At this stage, seeing no obvious difference from the position/velocity example, you might take Pauli's lead and be tempted to respond that there is no point in thinking about such issues. If you can't actually measure the spin about different axes, what is the point in wondering whether the particle nevertheless has a definite spin—clockwise versus counterclockwise—about each? Quantum mechanics, and physics more generally, is obliged only to account for features of the world that can be